

Beam Squint Assisted User Localization in Near-Field Integrated Sensing and Communications Systems

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Abstract—Integrated sensing and communication (ISAC) has been regarded as a key technology for 6G wireless communications, in which large-scale multiple input and multiple output (MIMO) array with higher and wider frequency bands will be adopted. However, recent studies show that the beam squint phenomenon can not be ignored in wideband MIMO system, which generally deteriorates the communications performance. In this paper, we find that with the aid of true-time-delay lines (TTDs), the range and trajectory of the beam squint in near-field communications systems can be freely controlled, and hence it is possible to reversely utilize the beam squint for user localization. We derive the trajectory equation for *near-field beam squint points* and design a way to control such trajectory. With the proposed design, beamforming from different subcarriers would purposely point to different angles and different distances, such that users from different positions would receive the maximum power at different subcarriers. Hence, one can simply localize multiple users from the beam squint effect in frequency domain, and thus reduce the beam sweeping overhead as compared to the conventional time domain beam search based approach. Furthermore, we utilize the phase difference of the maximum power subcarriers received by the user at different frequencies in several times beam sweeping to obtain a more accurate distance estimation result, ultimately realizing high accuracy and low beam sweeping overhead user localization. Simulation results demonstrate the effectiveness of the proposed schemes.

Index Terms—Near-field, integrated sensing and communication, user localization, beam squint, 6G communication, true-time-delay lines.

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I. INTRODUCTION

ISAC has attracted much research interest recently and is deemed as one of the key technologies for the sixth generation (6G) communications [1], [2], [3], [4]. One of the ideas is to use the communication signaling to sense the user's position as a simultaneous functionality. Once the user's position is obtained, the base station (BS) system can not only better serve the user's communication [5], but also supports a variety of intelligent services, such as path planning for the Internet of Vehicles, intrusion detection for intelligent security and unmanned aerial vehicle cooperative work [6], [7], [8].

When the user is far from the BS, say in the far-field area, the BS typically focuses on the users' angles in order to provide directional far-field communication beams in early research, which can be regarded as the estimation of the direction of arrival (DOA) problem. In the study of array signal processing, a number of classical DOA estimation methods were proposed. For example, multiple signal classification (MUSIC) algorithm can realize DOA estimation based on matrix eigenspace decomposition and is an important cornerstone of direction finding theory of spatial spectrum estimation [9]. In addition, DOA estimation methods based on compressed sensing have also been proposed that take the advantage of the sparse characteristics of signals [10], [11], [12]. Recently, Z. Chen proposed a RIS-based DOA estimation method for ISAC system [13]. P. Chen proposed a super-resolution DOA estimation method based on deep learning, and discussed the basic problem of DOA estimation in ISAC system [14].

As MIMO array becomes larger and larger in 5G and future 6G communications, the user's position may stay within the Rayleigh distance, resulting in the evolution from far-field effect to near-field effect of electromagnetic field [15], [16], [17]. The Rayleigh distance is usually expressed as $Z = \frac{2D^2}{\lambda}$, where λ is the wavelength and D is antenna array aperture. For example, the wavelength of a BS operating in 30 GHz is $\lambda = 0.01m$. Assuming the BS is configured with a uniform linear array (ULA) with 128 antenna elements and half wavelength antenna spacing, then the aperture of this BS is $D = \frac{(128-1) \times 0.01m}{2} = 0.635m$, and the near-field Rayleigh distance of this BS is about $Z = \frac{2D^2}{\lambda} = \frac{2 \times 0.635m^2}{0.01m} \approx 80m$. This distance is a typical working distance in the future smart

city, smart transportation and smart factory [18]. Thus we need to consider how to realize user localization in near-field ISAC system. There have been some interesting work in this area recently. Y. Jiang proposed a reconfigurable intelligence surface (RIS) assisted near-field communication and localization method [19]. J. Yang proposed a localization method based on extremely large lens antenna array, which utilizes the window effect for energy focusing property of ExLens [20].

However, these methods all consider narrowband signals, but 6G communication will adopt wider frequency bands as well as advanced beamforming to achieve higher rate communication transmission [21], [22], [23]. It has been shown in [24] that for a wideband communication system using orthogonal frequency division multiplexing (OFDM) modulation, the phenomenon of *beam squint* would appear in far-field communications, in which the beamforming from different subcarriers would point to different directions, making the energy of part subcarriers deviate from the desired user position. Beam squint effect is usually considered as a negative effect and should be mitigated with the aid of the true-time-delay lines (TTDs) [25], [26], [27]. For example, an efficient method was proposed to realize the wideband hybrid beamforming by using TTDs, which greatly alleviates the energy leakage caused by beam squint while considering the economic cost [28]. Nevertheless, it is recently reported that beam squint can be inversely used to sense the DOA for different users thanks to the squint beam over different subcarriers [29].

When the user moves to near-field area, the beam squint phenomenon will also occur. Unlike the plane wave assumption for the far-field, the spherical wave assumption is usually used for the near-field [30], [31], [32]. In this case, due to the propagation characteristics of the spherical waves, beamforming from different subcarriers will focus at different positions. This property is conducive for us to locate the positions of near-field users, but there is no related discussions in literature, to the best of the authors' knowledge.

In this paper, we fill this research gap and propose the low-overhead near-field user localization algorithms by reversely utilizing the wideband beam squint effect, in which the BS can use OFDM signals to quickly sense the users' positions in frequency domain. The contributions of this paper are summarized as follows.

- We derive the near-field wideband channel models in time domain and frequency domain as well as the corresponding beamforming models.
- We analyze the near-field beam squint phenomenon and derive the trajectory equation of *near-field beam squint points*. Then we design a way to control the range and trajectory of the beam squint points with the help of TTDs.
- We reversely utilize the controllable beam squint for user localization and propose a low complexity near-field user localization scheme based on TTDs-assisted beam squint.

- We propose a high accuracy localization scheme based on controllable beam squint, which utilizes the phase difference of maximum power subcarriers with different frequencies received by users in several times beam sweeping to realize more accurate distance estimation.

The remainder of this paper is organized as follows. In Section II, we introduce the system model of the near-field communications. In Section III, the near-field beam squint phenomenon is analyzed and we propose a controllable beam squint strategy with the help of TTDs. Then we proposed several near-field user localization schemes based on controllable beam squint in Section IV. Simulation results and conclusions are given in Section V and Section VI.

Notations: Lower-case and upper-case boldface letters $\mathbf{a}$ and $\mathbf{A}$ denote a vector and a matrix, respectively; $\mathbf{a}^T$ and $\mathbf{a}^H$ denote the transpose and the conjugate transpose of vector $\mathbf{a}$, respectively; $\text{diag}(\mathbf{a})$ denotes a diagonal matrix with the diagonal elements constructed from $\mathbf{a}$. $[\mathbf{a}]_n$ denotes the n -th element of the vector $\mathbf{a}$. Symbols $\mathbf{1}$ represent the all-ones matrix or vector. $\mathcal{R}\{\cdot\}$ and $\arg(\cdot)$ denote the real part and the phase of a complex number, a vector or a matrix, respectively. $\lceil x \rceil$ denotes the minimum integer that is not smaller than x . $\delta(\cdot)$ is the Dirac delta function, and $|\cdot|$ denotes the absolute operator.

II. SYSTEM MODEL

A wideband massive MIMO system working in mmWave frequency band with OFDM modulation is considered in this paper, in which the BS is equipped with an N -antenna ULA with antenna spacing d and a single radio frequency (RF) chain. The n -th antenna is deployed at $(0, nd)$, where $n = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$, and the antenna array aperture is $D = (N-1)d \approx Nd$. The carrier frequency and transmission bandwidth are denoted as f_c and W , and the subcarrier frequency range is $[f_c - \frac{W}{2}, f_c + \frac{W}{2}]$. Assuming there are a total of $M+1$ subcarriers, the 0-th subcarrier has the lowest frequency $f_0 = f_c - \frac{W}{2}$ and the m -th subcarrier frequency is $f_m = f_0 + m\frac{W}{M}$. Particularly, by denoting the baseband frequency of the m -th subcarrier as $\tilde{f}_m = m\frac{W}{M}$, there is $f_m = f_0 + \tilde{f}_m$, where $m = 0, 1, 2, \dots, M$. Assume that the BS serves K users, and each user is equipped with a single antenna. Suppose that the k -th user is located at (x_k, y_k) , and the corresponding polar coordinate is (r_k, θ_k) . Then the distance between the n -th antenna and the k -th user is $r_{k,n} = \sqrt{x_k^2 + (y_k - nd)^2}$.

According to [15], the boundary between near-field and far-field is determined by the Rayleigh distance $Z = \frac{2D^2}{\lambda}$, where λ is the wavelength. Here all users are assumed to be in the near-field range. For mmWave frequency band communications, we consider only one line-of-sight (LoS) channel between the user and the BS while the proposed study can be readily extended to the multipath scenario. Denote $\tau_{k,n}$ as the time delay of the path from the n -th antenna of the BS to the antenna of the k -th user. Under the near-field assumption, there is $\tau_{k,n} = \frac{r_{k,n}}{c}$, where c represents the speed of light. In the near-field area, the Fresnel approximation [33] is usually

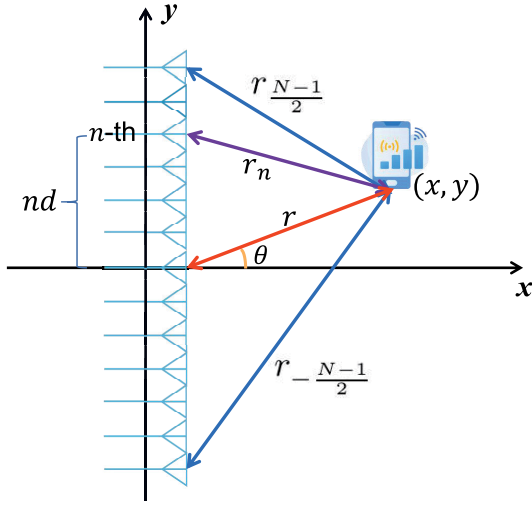


Fig. 1. Near-field system model.

employed to expand $r_{k,n}$ as

$$\begin{aligned} r_{k,n} &= \sqrt{x_k^2 + (y_k - nd)^2} = \sqrt{r_k^2 + n^2 d^2 - 2r_k n d \sin \theta_k} \\ &\approx r_k \left[1 + \frac{1}{2} \left(\frac{n^2 d^2}{r_k^2} - \frac{2nd \sin \theta_k}{r_k} \right) - \frac{1}{8} \left(\frac{n^2 d^2}{r_k^2} - \frac{2nd \sin \theta_k}{r_k} \right)^2 \right] \\ &\approx r_k - nd \sin \theta_k + \frac{n^2 d^2 \cos^2 \theta_k}{2r_k}. \end{aligned} \quad (1)$$

Suppose that the signal transmitted by the BS is $s(t)$, and the corresponding passband signal can be represented as $\mathcal{R}\{s(t)e^{j2\pi f_0 t}\}$. Then the signal received by the k -th user will be a delayed version of the transmit signal with amplitude attenuation. Specifically, the signal that the k -th user receives from the n -th antenna is $\mathcal{R}\{\alpha_{k,n}s(t - \tau_{k,n})e^{j2\pi f_0(t - \tau_{k,n})}\}$, where $\alpha_{k,n}$ is the channel path loss. Since it is a downlink channel, the k -th user eventually receives the signal as $\mathcal{R}\left\{\sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} \alpha_{k,n}s(t - \tau_{k,n})e^{j2\pi f_0(t - \tau_{k,n})}\right\}$. Then the time-domain downlink channel between the n -th antenna of the BS and the k -th user can be expressed as

$$h_{k,n}(t) = \alpha_{k,n}e^{-j2\pi f_0 \tau_{k,n}}\delta(t - \tau_{k,n}), \quad (2)$$

where $\delta(\cdot)$ denotes the Dirac delta function. By taking the Fourier transform of (2), the corresponding frequency-domain channel on the m -th subcarrier can be represented as

$$h_{k,n,m}^F = \alpha_{k,n,m}e^{-j2\pi f_m \tau_{k,n}}. \quad (3)$$

According to [34], the specific channel fading can be modeled as $\alpha_{k,n,m} = \frac{c}{4\pi f_m r_{k,n}}$. Because the distance between the user and the center of BS is generally much larger than the antenna array aperture, i.e., $r_k \gg D$, it is usually assumed that $\alpha_{k,-(N-1)/2,m} \approx \dots \approx \alpha_{k,(N-1)/2,m} \approx \alpha_{k,m} = \frac{c}{4\pi f_m r_k}$. Then the LoS channel vector of the k -th user $\mathbf{h}(r_k, \theta_k, f_m) \in \mathbb{C}^{N \times 1}$ can be modeled as

$$\begin{aligned} [\mathbf{h}(r_k, \theta_k, f_m)]_n &= \alpha_{k,m}e^{-j2\pi f_m \frac{r_{k,n}}{c}} \\ &= \alpha_{k,m}e^{-j2\pi f_m \frac{r_k - nd \sin \theta_k + \frac{n^2 d^2 \cos^2 \theta_k}{2r_k}}{c}}, \end{aligned} \quad (4)$$

where $[\mathbf{h}(r_k, \theta_k, f_m)]_n$ denotes the channel between the n -th antenna and the k -th user on the m -th subcarrier.

In order to provide better communication links for the users, BS adopts the phase-shifters (PSs) based beamforming for the single RF chain. We consider that the optimal near-field beamforming vector $\mathbf{w}(r_0, \theta_0) \in \mathbb{C}^{N \times 1}$ is determined by the point (r_0, θ_0) and the lowest carrier frequency f_0 , that is, $\mathbf{w}(r_0, \theta_0)$ should match the phase of $\mathbf{h}(r_0, \theta_0, f_0)$ and can be denoted as

$$[\mathbf{w}(r_0, \theta_0)]_n = \frac{1}{\sqrt{N}}e^{j \arg([\mathbf{h}(r_0, \theta_0, f_0)]_n)} = \frac{1}{\sqrt{N}}e^{-j2\pi f_0 \frac{r_{0,n}}{c}}, \quad (5)$$

where $\arg(x)$ denotes the phase of x . Then the downlink transmission process of the k -th user on the m -th subcarrier can be expressed as

$$y_{k,m} = \mathbf{h}_{k,m}^H \cdot \mathbf{w} \cdot s_{k,m} + n_{k,m}, \quad (6)$$

where $y_{k,m} \in \mathbb{C}^{1 \times 1}$ denotes the complex signal received by the k -th user on the m -th subcarrier. Moreover, $\mathbf{h}_{k,m} = \mathbf{h}(r_k, \theta_k, f_m)$ is the channel vector, $s_{k,m}$ is the data transmitted by the m -th subcarrier, and $n_{k,m}$ is the zero-mean additive complex Gaussian white noise with variance of σ_k^2 . Then the complex signal received by the k -th user at each subcarrier can be formed into a vector $\mathbf{y}_k \in \mathbb{C}^{(M+1) \times 1}$ as

$$\mathbf{y}_k = [y_{k,0}, y_{k,1}, \dots, y_{k,M}]^T = \text{diag}(\mathbf{H}_k^H \cdot \mathbf{w}) \cdot \mathbf{s}_k + \mathbf{n}_k, \quad (7)$$

where $\mathbf{H}_k \triangleq [\mathbf{h}_{k,0}, \mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,M}] \in \mathbb{C}^{N \times (M+1)}$ represents the overall downlink channel at different subcarriers in frequency domain, $\mathbf{s}_k \triangleq [s_{k,0}, s_{k,1}, \dots, s_{k,M}]^T \in \mathbb{C}^{(M+1) \times 1}$ represents the data information carried by all subcarriers, and $\mathbf{n}_k \triangleq [n_{k,0}, n_{k,1}, \dots, n_{k,M}]^T \in \mathbb{C}^{(M+1) \times 1}$ represents the additive noise vector.

III. NEAR-FIELD BEAM SQUINT

A. BS View: Trajectory of Near-Field Beam Squint Points

In order to concentrate the beam transmitted by the BS at the desired near-field position (r_0, θ_0) , under the limitations of only one single RF and the PSs structure, the BS needs to employ the near-field beamforming vector $\mathbf{w}(r_0, \theta_0)$ for all the subcarriers. In narrowband systems, due to the small frequency difference between different subcarriers, all the subcarriers will be approximately focused at the desired position (r_0, θ_0) . However, with wideband communications, i.e., W is large, the beamforming of the m -th subcarrier may not point to the desired position (r_0, θ_0) . Such a phenomenon is named as *near-field beam squint* as the near-field beamforming position gradually “squint” over the frequency.

Specifically, based on formula (4), the near-field array response vector at any position (r, θ) on the m -th subcarrier can be defined as $\mathbf{b}(r, \theta, f_m) \in \mathbb{C}^{N \times 1}$, which satisfies $[\mathbf{b}(r, \theta, f_m)]_n = e^{-j2\pi f_m \frac{r_n}{c}}$. Moreover, $r_n = \sqrt{x^2 + (y - nd)^2}$ denotes the distance between the near-field point (r, θ) and the n -th antenna of BS. When $\mathbf{w}(r_0, \theta_0)$ is designed by the desired position (r_0, θ_0) and frequency f_0 ,

the array gain at any position (r, θ) on the m -th subcarrier can be calculated as

$$g(r, \theta, f_m, r_0, \theta_0) = |\mathbf{w}^H(r_0, \theta_0) \cdot \mathbf{b}(r, \theta, f_m)|$$

$$= \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi f_0 \frac{r_{0,n}}{c}} e^{-j2\pi f_m \frac{r_n}{c}} \right|, \quad (8)$$

where $r_{0,n} = \sqrt{x_0^2 + (y_0 - nd)^2}$ denotes the distance between the desired point (r_0, θ_0) and the n -th antenna of BS. Then we can apply the Fresnel approximation as (1) for both $r_{0,n}$ and r_n in (8) and obtain

$$g(r, \theta, f_m, r_0, \theta_0)$$

$$= \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j\frac{2\pi n^2 d^2}{c} (f_m \frac{\cos^2 \theta}{2r} - f_0 \frac{\cos^2 \theta_0}{2r_0})} e^{-j\frac{2\pi n d}{c} (f_m \sin \theta - f_0 \sin \theta_0)} \right|. \quad (9)$$

The maximum value of the array gain $g(x, y, f_m, x_0, y_0)$ is $\sqrt{N}$. When $m = 0$, g is obviously maximized at the position $(r, \theta) = (r_0, \theta_0)$. However, for the other subcarriers with frequency f_m , $m \neq 0$, the position (r, θ) that maximizes g is not (r_0, θ_0) , and will be defined as (r_m, θ_m) , i.e., the beamforming focus squint to other position. We call (r_m, θ_m) as *near-field beam squint points* in this paper, which satisfies

$$(r_m, \theta_m)$$

$$= \underset{(r, \theta)}{\operatorname{argmax}} \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j\frac{2\pi n^2 d^2}{c} \chi_{1,m}} e^{-j\frac{2\pi n d}{c} \chi_{2,m}} \right|$$

$$= \underset{(r, \theta)}{\operatorname{argmax}} \frac{1}{\sqrt{N}} F\left(\frac{2\pi d^2}{c} \chi_{1,m}, \frac{-2\pi d}{c} \chi_{2,m}\right), \quad (10)$$

where $\chi_{1,m} = f_m \frac{\cos^2 \theta}{2r} - f_0 \frac{\cos^2 \theta_0}{2r_0}$, $\chi_{2,m} = f_m \sin \theta - f_0 \sin \theta_0$, and the binary function $F(x, y)$ is defined as $F(x, y) = |\sum_{n=-(N-1)/2}^{(N-1)/2} e^{jn^2 x} e^{jny}|$. Since $F(\frac{2\pi d^2}{c} (f_m \frac{\cos^2 \theta}{2r} - f_0 \frac{\cos^2 \theta_0}{2r_0}), \frac{-2\pi d}{c} (f_m \sin \theta - f_0 \sin \theta_0)) \leq N$ and $F(0, 0) = N$, considering the finite range of the near-field, a feasible solution for (r_m, θ_m) is

$$\sin \theta_m = \frac{f_0}{f_m} \sin \theta_0, \quad (11)$$

$$r_m = r_0 \cdot \frac{f_m}{f_0} \cdot \frac{\cos^2 \theta_m}{\cos^2 \theta_0}, \quad (12)$$

where $\cos^2 \theta_m = 1 - \sin^2 \theta_m = 1 - (\frac{f_0}{f_m} \sin \theta_0)^2$.

Hence the angle and distance of the near-field beam squint point of the m -th subcarrier is determined by (11) and (12), respectively. An example is presented in Fig. 2, which clearly shows the case of the near-field beam squint from the BS view. It is seen that as the subcarrier frequency f_m increases, the beamforming from different subcarriers would focus on different positions and can be connected into a curve trajectory. The start point of the trajectory is determined by the 0-th subcarrier with frequency f_0 while the end point of the trajectory is determined by the M -th subcarrier with frequency f_M . In Fig. 2, the lowest frequency is $f_0 = 30$ GHz whose beamforming focuses at $(10m, 60^\circ)$, while the highest frequency

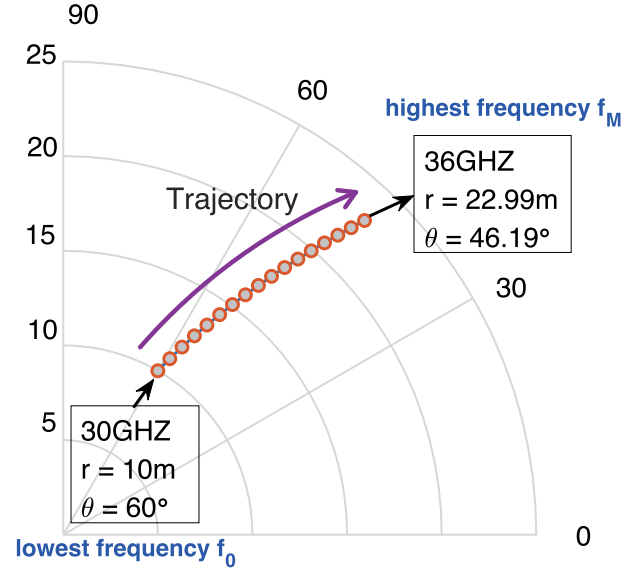


Fig. 2. Trajectory of near-field beam squint points, where $f_0 = 30$ GHz, $W = 6$ GHz, $M = 16$, the beamforming of the subcarrier with the lowest frequency f_0 is set to focus on $(10m, 60^\circ)$.

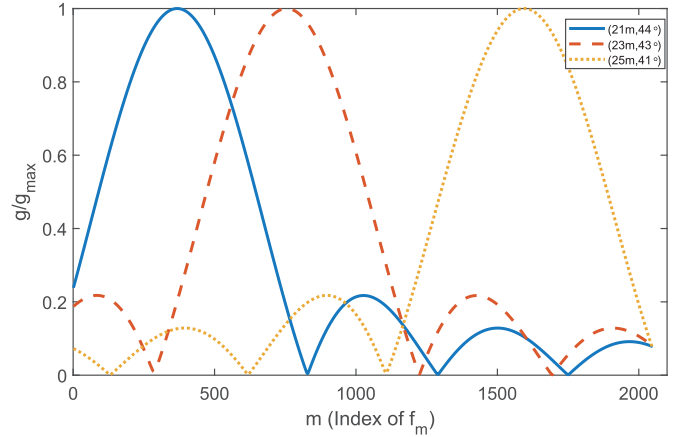


Fig. 3. The normalized power spectra of received signals of different users under natural beam squint, where the users are located at $(21m, 44^\circ)$, $(23m, 43^\circ)$ and $(25m, 41^\circ)$.

subcarrier is 36 GHz whose beamforming eventually squints to $(22.99m, 46.19^\circ)$. Clearly, near-field beam squint cannot be ignored in the wideband ISAC system.

B. Users View: Power Spectrum of All Subcarriers

As shown in formula (7), for the k -th user, the complex received signal at different subcarriers can constitute a vector $\mathbf{y}_k$. By taking the modulus of each element in $\mathbf{y}_k$, the power of the k -th user's received signal at different frequencies can be obtained as

$$\mathbf{g}_k = [|y_{k,0}|, |y_{k,1}|, \dots, |y_{k,M}|]^T, \quad (13)$$

where the specific expression for $y_{k,m}$ is given by (6). Taking the index of the subcarrier m as the horizontal axis, and taking the value of $|y_{k,m}|$ as the vertical axis, we can draw the curve of the power spectrum of the k -th user's received signal.

An example is shown in Fig. 3, in which the lowest frequency f_0 is 30 GHz, the highest frequency f_M is 33 GHz, and the number of subcarriers is $M + 1 = 2048$. By adjusting

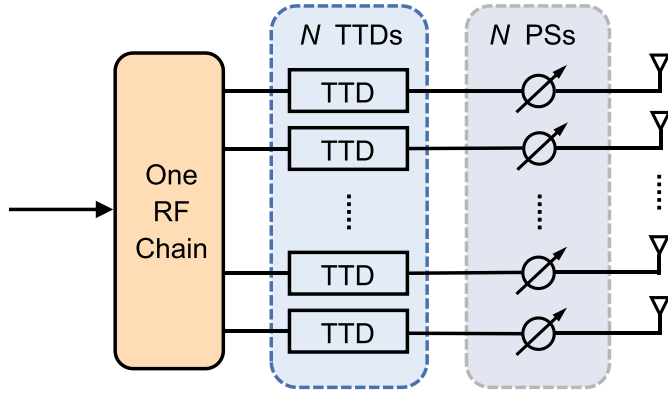


Fig. 4. Schematic diagram of the proposed TTDs assisted BS link.

the PSs such that the beamforming of the lowest frequency subcarrier points to $(20m, 45^\circ)$, then the beamforming of the highest frequency subcarrier will squint to $(25.82m, 40.00^\circ)$. We plot the received power spectrum of three different users located within the range of $\{(r, \theta) | 20m \leq r \leq 25.82m, 40^\circ \leq \theta \leq 45^\circ\}$ in Fig. 3. It is clearly seen that the power spectrums of the users at different positions are different, which will help BS to sense the locations of the users according to their special power spectrums. By doing this, one may reduce the time of beam sweeping as compared to the narrowband system, i.e., the beam squint effect surprisingly has the positive effect when sensing the users' positions. Nevertheless, the range of near-field beam squint mainly depends on the relative transmission bandwidth of the system, and it is difficult to cover the entire required sensing range within only one OFDM symbol. Therefore, we must consider further expanding the range of beam squint to realize efficient sensing.

C. BS View: Controllable Beam Squint Based on TTDs

Different from a PS that generates a fixed phase over the whole frequency band, a TTD line can provide a programmable true time delay that induces the varying phase over frequencies [35], [36]. As shown in Fig. 4, we assume that each phase shifter is cascaded with one TTD line [29]. The response of the n -th phase shifter can be expressed as $e^{-j2\pi\phi_n}$, where ϕ_n denotes the phase shift amount of the n -th PS. The time domain response of the n -th TTD can be expressed as $\delta(t - t_n)$, and the corresponding frequency domain response is $e^{-j2\pi\tilde{f}t_n}$, where t_n denotes the time delay amount of the n -th TTD and $\tilde{f}$ is the baseband frequency. Hence the new array beamforming vector assisted by TTDs can be expressed as $\tilde{\mathbf{w}} \in \mathbb{C}^{N \times 1}$, and

$$[\tilde{\mathbf{w}}]_n = \frac{1}{\sqrt{N}} e^{-j2\pi\phi_n} e^{-j2\pi\tilde{f}t_n}. \quad (14)$$

With TTDs, the array gain at any near-field position (r, θ) on the m -th subcarrier is

$$\begin{aligned} \tilde{g}(r, \theta, f_m, \tilde{\mathbf{w}}) &= |\tilde{\mathbf{w}}^H \cdot \mathbf{b}(r, \theta, f_m)| \\ &= \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi\phi_n} e^{j2\pi\tilde{f}t_n} e^{-j2\pi f_m \frac{r_n}{c}} \right|. \end{aligned} \quad (15)$$

Similarly, $\tilde{g}(r, \theta, f_m, \tilde{\mathbf{w}}) \leq \sqrt{N}$ and one of the solutions to maximize $\tilde{g}(r, \theta, f_m, \tilde{\mathbf{w}})$ need the following equation to be satisfied

$$f_m \frac{r_n}{c} - \phi_n - \tilde{f} t_n = p_{m,n}, \quad (16)$$

where $p_{m,n}$ is a set of integers introduced to align the phases in (15), and equation (16) should hold for all the $n = -\frac{N-1}{2}, \dots, \frac{N-1}{2}$. In addition, we mark the near-field beam squint point of the m -th subcarrier assisted by TTDs as $(\tilde{r}_m, \tilde{\theta}_m)$, i.e.,

$$(\tilde{r}_m, \tilde{\theta}_m) = \underset{(r, \theta)}{\operatorname{argmax}} \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi\phi_n} e^{j2\pi\tilde{f}t_n} e^{-j2\pi f_m \frac{r_n}{c}} \right|. \quad (17)$$

Although the start point and the end point of the beam squint trajectory are determined by subcarrier f_0 and f_M respectively, we can control the start point and the end point of the beam squint trajectory with the help of TTDs. By adjusting the values of the PSs, we can let the beamforming of the 0-th subcarrier with frequency f_0 point to the start point (r_0, θ_0) . Specifically, we set $f_m = f_0$, $\tilde{f}_m = \tilde{f}_0 = 0$, $r_n = r_{0,n}$, $p_{m,n} = 0$ in (16), and derive $\phi_n = \frac{f_0 r_{0,n}}{c}$. Then we set the phase shift of the n -th PS as ϕ_n . By adjusting the values of the TTDs, we can let the beamforming of the M -th subcarrier with frequency f_M point to the end point (r_c, θ_c) . Specifically, we set $f_m = f_M$, $\tilde{f}_m = \tilde{f}_M = W$, $r_n = r_{c,n}$, $p_{m,n} = 0$ in (16), and derive $t_n = \frac{f_M}{Wc} r_{c,n} - \frac{\phi_n}{W}$. Then we set the time delay value of the n -th TTD as t_n .

Afterward, the beams from the 0-th subcarrier to the M -th subcarrier will squint from the start point (r_0, θ_0) to the end point (r_c, θ_c) . By substituting $\phi_n = \frac{f_0 r_{0,n}}{c}$ and $t_n = \frac{f_M}{Wc} r_{c,n} - \frac{\phi_n}{W}$ in (17), the near-field controllable beam squint point of the m -th subcarrier $(\tilde{r}_m, \tilde{\theta}_m)$ satisfies

$$\begin{aligned} (\tilde{r}_m, \tilde{\theta}_m) &= \underset{(r, \theta)}{\operatorname{argmax}} \frac{1}{\sqrt{N}} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j \frac{2\pi n^2 d^2}{c^2} \beta_{1,m}} e^{-j \frac{2\pi n d}{c} \beta_{2,m}} \right| \\ &= \underset{(r, \theta)}{\operatorname{argmax}} \frac{F\left(\frac{2\pi d^2}{c^2} \beta_{1,m}, -\frac{2\pi d}{c} \beta_{2,m}\right)}{\sqrt{N}}, \end{aligned} \quad (18)$$

where $\beta_{1,m} = f_m \frac{\cos^2 \theta}{2r} - \frac{(W - \tilde{f}_m) f_0}{W} \frac{\cos^2 \theta_0}{2r_0} - \frac{f_M \tilde{f}_m}{W} \frac{\cos^2 \theta_c}{2r_c}$ and $\beta_{2,m} = f_m \sin \theta - \frac{(W - \tilde{f}_m) f_0}{W} \sin \theta_0 - \frac{f_M \tilde{f}_m}{W} \sin \theta_c$. Since $F\left(\frac{2\pi d^2}{c^2} \beta_{1,m}, -\frac{2\pi d}{c} \beta_{2,m}\right) \leq N$ and $F(0, 0) = N$, considering the finite range of near-field area, a feasible solution for $(\tilde{r}_m, \tilde{\theta}_m)$ is

$$\begin{aligned} \sin \tilde{\theta}_m &= \frac{(W - \tilde{f}_m) f_0}{W f_m} \sin \theta_0 + \frac{(W + f_0) \tilde{f}_m}{W f_m} \sin \theta_c, \quad (19) \\ \frac{1}{\tilde{r}_m} &= \frac{1}{r_0} \frac{(W - \tilde{f}_m) f_0}{W f_m} \frac{\cos^2 \theta_0}{\cos^2 \tilde{\theta}_m} + \frac{1}{r_c} \frac{(W + f_0) \tilde{f}_m}{W f_m} \frac{\cos^2 \theta_c}{\cos^2 \tilde{\theta}_m}. \end{aligned} \quad (20)$$

In summary, from the view of the BS, we can autonomously control the start point and the end point of the squinted beams with the help of TTDs, such that different subcarriers will focus at different positions in the way we would expect. Some examples of different control schemes are shown in

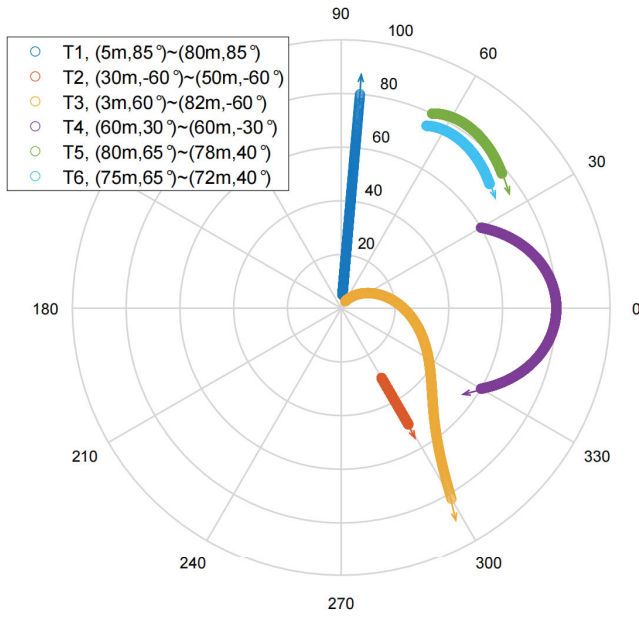


Fig. 5. TTDs-assisted near-field controllable beam squint trajectories.

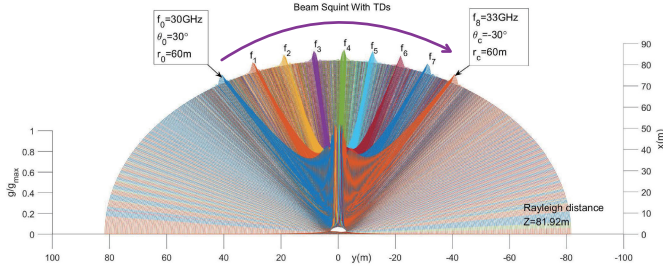

 Fig. 6. Trajectory of TTDs-assisted beam squint. Where $f_0 = 30$ GHz, $W = 3$ GHz, $M = 8$, the lowest frequency carrier f_0 is set to focus on $(60m, 30^\circ)$ and the highest frequency carrier f_M is set to focus on $(60m, -30^\circ)$.

Fig. 5, where $M = 2048$ and $N = 128$. Trajectory T1 and T2 are two straight lines in the radial directions, where T1 squints from $(5m, 85^\circ)$ to $(80m, 85^\circ)$ while T2 squints from $(30m, -60^\circ)$ to $(50m, -60^\circ)$. If we already know the user's angle, then using the radius trajectories like T1 or T2 allows to quickly sense the user's distance. Trajectory T3 squints from $(3m, 60^\circ)$ to $(82m, -60^\circ)$, and the range of squint spans the entire expected sensing angle and distance regions. Trajectory T4 squints from $(60m, 30^\circ)$ to $(60m, -30^\circ)$ with $t_{n,T4} \in [0.1889\mu s, 0.2112\mu s]$ in a symmetrical form. This symmetry is more conducive to obtaining the user's angle. Trajectories T5 and T6 show the controllable beam squint in a smaller range, which facilitates beam sweeping in a known narrow area.

Let us now show how the trajectory T4 can obtain all users' angles. We let $M = 8$ to clearly show the beamforming from different subcarriers. The power of the received signal of each subcarrier at all positions can be calculated. Fig. 6 shows the normalized received power of all discrete field points through the Electromagnetic distance $3.17m$ to Rayleigh distance $81.92m$ by distance step $0.4m$ and through angle range 90° to -90° by angle step -0.5° . As the frequency

gradually increases, the subcarrier squints from 30° to -30° , and different subcarriers have peaks in different directions. With this feature, we can determine the user's angle according to the subcarrier frequency of the maximum power fed back by the user, which will be densely effective when the number of subcarriers $M + 1$ is large enough.

IV. NEAR-FIELD USER LOCALIZATION SCHEME WITH CONTROLLABLE BEAM SQUINT

A. Baseline: Traditional Beam Training Based Near-Field User Localization Scheme

Beam training is an effective method for obtaining channel state information (CSI) in current communication systems. Through the beam training process between BS and users, the best beamforming vector can be selected from a set of pre-established near-field codebook. Due to the uniqueness of each codeword in the codebook, this process can also estimate the position of near-field users simultaneously. Assume that the sensing range required by the BS is $\{(r, \theta) | r_{min} \leq r \leq r_{max}, \theta_{min} \leq \theta \leq \theta_{max}\}$, and a traditional beam training based localization process can be divided into two stages: angle estimation and distance estimation.

During the angle estimation stage, the angle sensing range can be uniformly divided into I^a discrete angle values, forming I^a discrete near-field angle units. The i^a -th unit can be represented as $(r_a, \theta_{i^a}) = (r_a, \theta_{min} + (i^a - 1)\Delta\theta)$, where $\Delta\theta = \frac{\theta_{max} - \theta_{min}}{I^a - 1}$ and $i^a = 1, 2, \dots, I^a$. Then BS needs to implement I^a times beam sweeping based on PSs for all these units, and the beamforming vector in the i^a -th beam sweeping is $\mathbf{w}(r_a, \theta_{i^a})$. Assuming that BS emits all 1 pilot signals, the received power of the k -th user on the m -th subcarrier can be denoted as $g_{k,m,i^a} = |\mathbf{h}_{k,m}^H \mathbf{w}(r_a, \theta_{i^a}) + n_{k,m,i^a}|$. Then the angle of the k -th user can be estimated as $\hat{\theta}_k^{Base} = \arg\max_{\theta_{i^a}} \sum_{m=0}^M g_{k,m,i^a}$. In order to further estimate the distance of the k -th user, the BS divides the distance sensing range at $\hat{\theta}_k^{Base}$ direction into I^d discrete distance values, forming I^d near-field distance units. The $i^{d,k}$ -th unit is $(r_{i^d}, \hat{\theta}_k^{Base}) = (r_{min} + (i^{d,k} - 1)\Delta r, \hat{\theta}_k^{Base})$, where $\Delta r = \frac{r_{max} - r_{min}}{I^d - 1}$ and $i^{d,k} = 1, 2, \dots, I^d$. Similar to the angle estimation process, the distance of the k -th user can be estimated as $\hat{r}_k^{Base} = \arg\max_{r_{i^d}} \sum_{m=0}^M |\mathbf{h}_{k,m}^H \mathbf{w}(r_{i^d}, \hat{\theta}_k^{Base}) + n_{k,m,i^d}|$.

For ease of expression, this beam training based scheme will be abbreviated as *TBT-Baseline* in this paper. Due to the fact that this traditional scheme can only concentrate the beam on one near-field position in one time beam sweeping, the TBT-Baseline scheme requires a total of $I = I^a + K \times I^d$ times beam sweeping to obtain the positions of all K users.

B. Proposed: A Low Complexity User Localization Scheme Based on Controllable Beam Squint

With the help of TTDs, the BS can adjust the values of the PSs as $\phi_n = \frac{f_0 r_{0,n}}{c}$ and the values of the TTDs as $t_n = \frac{f_M}{W_c} r_{c,n} - \frac{\phi_n}{W}$ to make the beamforming of the subcarriers from the 0-th subcarrier to the M -th subcarrier squint from the start

position (r_0, θ_0) to the end position (r_c, θ_c) . Assuming the BS emits all 1 pilot signals, under ideal noiseless conditions, the complex received signal of the k -th user on the m -th subcarrier now can be represented as

$$\begin{aligned} \tilde{y}_{k,m} &= \mathbf{h}^H(r_k, \theta_k, f_m) \cdot \tilde{\mathbf{w}} \\ &= \frac{\alpha_{k,m}}{\sqrt{N}} \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi f_m \frac{r_{k,n}}{c}} e^{-j2\pi \phi_n} e^{-j2\pi \tilde{f}_m t_n}. \end{aligned} \quad (21)$$

For the convenience of analysis, we multiply $\tilde{y}_{k,m}$ by $\frac{4\pi\sqrt{N}}{\lambda_m}$ to eliminate the influence of different subcarriers in $\alpha_{k,m}$ and obtain the received signal after being processed as

$$\begin{aligned} \check{y}_{k,m} &= \frac{4\pi\sqrt{N}}{\lambda_m} \tilde{y}_{k,m} \\ &= \frac{1}{r_k} \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi \frac{f_m}{c} r_{k,n}} e^{-j2\pi \frac{(W-\tilde{f}_m)f_0}{Wc} r_{0,n}} e^{-j2\pi \frac{\tilde{f}_m f_M}{Wc} r_{c,n}} \\ &= \frac{1}{r_k} e^{j\xi_{0,k,m}} \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m}} e^{jn^2\xi_{2,k,m}}, \end{aligned} \quad (22)$$

where $\xi_{0,k,m} = \frac{2\pi}{Wc}[Wf_mr_k - (W - \tilde{f}_m)f_0r_0 - \tilde{f}_mf_Mr_c]$, $\xi_{1,k,m} = -\frac{2\pi d}{Wc}[Wf_m \sin \theta_k - (W - \tilde{f}_m)f_0 \sin \theta_0 - \tilde{f}_mf_M \sin \theta_c]$ and $\xi_{2,k,m} = \frac{2\pi d^2}{Wc}[Wf_m \frac{\cos^2 \theta_k}{2r_k} - (W - \tilde{f}_m)f_0 \frac{\cos^2 \theta_0}{2r_0} - \tilde{f}_mf_M \frac{\cos^2 \theta_c}{2r_c}]$.

Assume that the sensing range required by the BS is still $\{(r, \theta) | r_{min} \leq r \leq r_{max}, \theta_{min} \leq \theta \leq \theta_{max}\}$. The proposed low complexity localization scheme includes two stages: angle sensing and distance sensing. In the angle sensing stage, assume that $r_{mid1}, r_{mid2} \in [r_{min}, r_{max}]$ are two appropriate specific distance values. We target to use one time beam sweeping to get all K users' angle estimation results $\hat{\theta}_k^{Pro1}$, where $k = 1, 2, \dots, K$. The specific steps are as follows:

Firstly, by adjusting the values of PSs and TTDs, the beamforming of the 0-th and M -th subcarriers can be controlled to focus on positions (r_{mid1}, θ_{max}) and (r_{mid2}, θ_{min}) , respectively. Secondly, the BS simultaneously transmits over $M+1$ subcarriers, whose angles squint from θ_{max} to θ_{min} and distances squint from r_{mid1} to r_{mid2} , covering the whole space in the form of a curve, in which the trajectory is similar to trajectory T4 in Fig. 5. Thirdly, all users will receive $M+1$ subcarriers, and the power of the m -th subcarrier received by the k -th user is $\check{g}_{k,m} = |\check{y}_{k,m}| = \frac{1}{r_k} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m}} e^{jn^2\xi_{2,k,m}} \right|$. Substituting $r_0 = r_{mid1}$, $\theta_0 = \theta_{max}$, $r_c = r_{mid2}$ and $\theta_c = \theta_{min}$ into $\xi_{1,k,m}$ and $\xi_{2,k,m}$, we can obtain $\xi_{1,k,m}^{angle} = -\frac{2\pi d}{Wc}[Wf_m \sin \theta_k - (W - \tilde{f}_m)f_0 \sin \theta_{max} - \tilde{f}_mf_M \sin \theta_{min}]$ and $\xi_{2,k,m}^{angle} = \frac{2\pi d^2}{Wc}[Wf_m \frac{\cos^2 \theta_k}{2r_k} - (W - \tilde{f}_m)f_0 \frac{\cos^2 \theta_{max}}{2r_{mid1}} - \tilde{f}_mf_M \frac{\cos^2 \theta_{min}}{2r_{mid2}}]$. In order to simplify the second-order calculation of the near-field, we have to select appropriate r_{mid1} and r_{mid2} , such that there is $\xi_{2,k,m}^{angle} \approx p_{k,m}^{angle} \cdot 2\pi$ within the sensing angle range, where $p_{k,m}^{angle}$ is an integer introduced for approximate

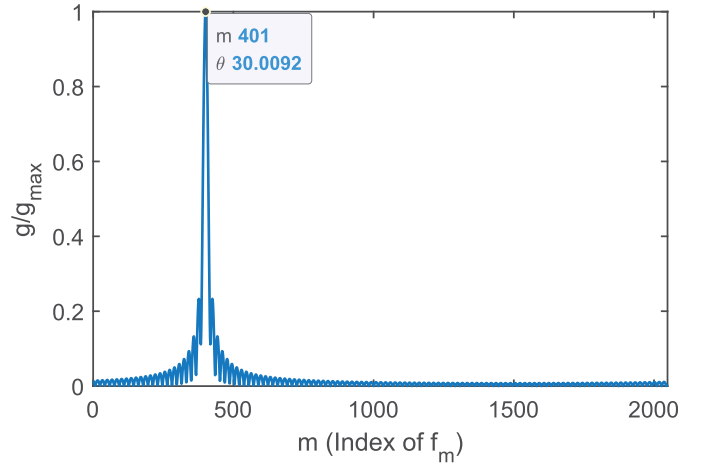


Fig. 7. Normalized power spectrum of received signal of the user at $(30m, 30^\circ)$ in angle sensing stage of the proposed low complexity localization scheme. $M = 2048$ and $N = 128$.

phase alignment. Then $\check{g}_{k,m}^{angle}$ can be reduced to

$$\check{g}_{k,m}^{angle} = \frac{1}{r_k} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m}^{angle}} \right| = \frac{1}{r_k} \left| \frac{\sin(\frac{\xi_{1,k,m}^{angle} N}{2})}{\sin(\frac{\xi_{1,k,m}^{angle}}{2})} \right|. \quad (23)$$

Since $\xi_{1,k,m}^{angle}$ is a linear function of f_m and the product of its two endpoints is $\xi_{1,k,0}^{angle} \cdot \xi_{1,k,M}^{angle} = \frac{4\pi^2 d^2 f_0 f_M}{c^2} (\sin \theta_k - \sin \theta_{max})(\sin \theta_k - \sin \theta_{min}) < 0$, when $M+1$ is large enough, there must be an approximate $f_{m_k}^{angle}$ to make $\xi_{1,k,m_k}^{angle} = 0$. This means that there will be a peak value $\check{g}_{k,m_k}^{angle}$ in the revised power spectrum of the k -th user $\check{\mathbf{g}}_k^{angle} = [\check{g}_{k,0}^{angle}, \check{g}_{k,1}^{angle}, \dots, \check{g}_{k,M}^{angle}]^T$. We call this subcarrier as the *maximum power subcarrier* corresponding to the k -th user in angle sensing stage, and record its frequency as $f_{k,d_\theta}^{Pro1} = f_{m_k}^{angle}$. Then the k -th user can feed back f_{k,d_θ}^{Pro1} to the BS, and the BS can calculate the angle estimation of the k -th user as $\hat{\theta}_k^{Pro1}$ by substituting $\hat{\theta}_m = \hat{\theta}_k^{Pro1}$, $f_m = f_{k,d_\theta}^{Pro1}$, $\tilde{f}_m = f_{k,d_\theta}^{Pro1} - f_0$, $\theta_0 = \theta_{max}$ and $\theta_c = \theta_{min}$ into (19). By doing so, we can obtain the angle estimation results for each user.

Although we make some approximations when calculating (23), Fig. 7 shows the simulation results using formula (22) without approximation. When the user is at $(30m, 30^\circ)$, we can get that the 401-th subcarrier is the maximum power subcarrier corresponding to this user in angle sensing stage. Then the BS can calculate the user's angle as $\hat{\theta}_{example}^{Pro1} = 30.0092^\circ$ from $f_{example,d_\theta}^{Pro1} = f_{401}$.

It should be noted that if more than two users are in the same angle direction, then their angle estimates will be the same. Hence the angle sensing stage will get $\check{K}$ different angle estimates, with $\check{K} \leq K$ and $\check{K} \leq M+1$. Let us mark these $\check{K}$ different angles as $\check{\theta}_1, \check{\theta}_2, \dots, \check{\theta}_{\check{K}}$. Then the BS needs to perform $\check{K}$ times of TTDs-assisted controllable beam squint to complete the distance sensing for all users. Assuming that $\hat{\theta}_k^{Pro1} = \check{\theta}_{\check{k}}$, we next sense the k -th user's distance $\hat{r}_k^{Pro1}$ in the $\check{k}$ -th beam sweeping with the following steps:

Firstly, by adjusting the values of PSs and TTDs, the beamforming of the 0-th and M -th subcarriers can be controlled

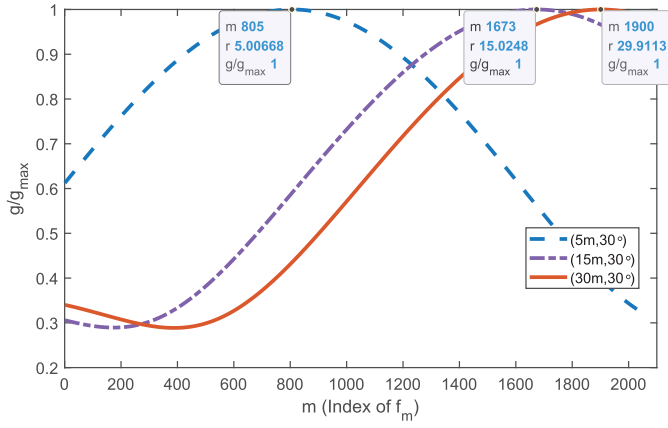


Fig. 8. Normalized power spectra of received signals of the users at $(5m, 30^\circ)$, $(15m, 30^\circ)$ and $(30m, 30^\circ)$ position in distance sensing stage of the proposed low complexity localization scheme. $M = 2048$ and $N = 128$.

to focus on $(r_{min}, \hat{\theta}_k^{Pro1})$ and $(r_{max}, \hat{\theta}_k^{Pro1})$, respectively. Secondly, the BS simultaneously transmits over $M+1$ subcarriers, whose beamforming angles are $\hat{\theta}_k^{Pro1}$, but the distances squint from r_{min} to r_{max} , covering the whole space in the form of a straight segment, which is similar to trajectory T1 in Fig. 5. Thirdly, the k -th user will receive $M+1$ subcarriers, and the power of the m -th subcarrier received by the k -th user is $\check{g}_{k,m}^{distance} = |\check{y}_{k,m}^{distance}| = \frac{1}{r_k} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m}^{distance}} e^{jn^2\xi_{2,k,m}^{distance}} \right|$. Substituting $\theta_k = \hat{\theta}_k^{Pro1}$, $r_0 = r_{min}$, $\theta_0 = \hat{\theta}_k^{Pro1}$, $r_c = r_{max}$ and $\theta_c = \hat{\theta}_k^{Pro1}$ into $\xi_{1,k,m}$ and $\xi_{2,k,m}$, we can obtain $\xi_{1,k,m}^{distance} = 0$ and $\xi_{2,k,m}^{distance} = \frac{\pi d^2 \cos^2 \hat{\theta}_k^{Pro1}}{Wc} [Wf_m \frac{1}{r_k} - (W - \tilde{f}_m) f_0 \frac{1}{r_{min}} - \tilde{f}_m f_M \frac{1}{r_{max}}]$. Then $\check{g}_{k,m}^{distance}$ can be reduced to

$$\check{g}_{k,m}^{distance} = \frac{1}{r_k} \left| \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn^2\xi_{2,k,m}^{distance}} \right|. \quad (24)$$

Since $\xi_{2,k,m}^{distance}$ is a linear function of f_m and the product of its two endpoints is $\xi_{2,k,0}^{distance} \cdot \xi_{2,k,M}^{distance} = \frac{\pi^2 d^4 \cos^4 \hat{\theta}_k^{Pro1} f_0 f_M}{c^2} \left(\frac{1}{r_k} - \frac{1}{r_{min}} \right) \left(\frac{1}{r_k} - \frac{1}{r_{max}} \right) < 0$, when $M+1$ is large enough, there must be an approximate $f_{m_k}^{distance}$ to make $\xi_{2,k,m_k}^{distance} = 0$. This means that there will be a peak value $\check{g}_{k,m_k}^{distance}$ in the revised power spectrum of the k -th user $\check{\mathbf{g}}_k^{distance} = [\check{g}_{k,0}^{distance}, \check{g}_{k,1}^{distance}, \dots, \check{g}_{k,M}^{distance}]^T$. We record the frequency of the maximum power subcarrier corresponding to the k -th user in the distance sensing stage as $f_{k,d_r}^{Pro1} = f_{m_k}^{distance}$. After the k -th user provides f_{k,d_r}^{Pro1} feedback to the BS, the BS can calculate the distance estimation of the k -th user as $\hat{r}_k^{Pro1}$ by substituting $\tilde{r}_m = \hat{r}_k^{Pro1}$, $f_m = f_{k,d_r}^{Pro1}$, $\tilde{f}_m = f_{k,d_r}^{Pro1} - f_0$, $r_0 = r_{min}$, $r_c = r_{max}$, $\hat{\theta}_m = \hat{\theta}_k^{Pro1}$ and $\theta_0 = \theta_c = \hat{\theta}_k^{Pro1}$ into (20). By doing so, we can obtain the distance estimation results for each user.

An example of distance sensing is shown in Fig. 8. For the user at $(30m, 30^\circ)$, we can get that the 1900-th subcarrier has the maximum power, and then calculate the user's distance as $\hat{r}_{example}^{Pro1} = 29.9113m$ from $f_{example,d_r}^{Pro1} = f_{1900}$. Furthermore, the normalized power spectrums of another two users in

direction 30° are also given in Fig. 8. We can see that the users at different distances receive the maximum power at different subcarriers, which allows BS to determine the user's distance. However, note that the power change of different subcarriers near the maximum power is flat, which is very different from the angle sensing as shown in Fig. 7. This means that the distance sensing in the proposed low complexity localization scheme is susceptible to the noise interference in environments with low signal-to-noise (SNR) ratio.

C. Proposed: A High Accuracy User Localization Scheme Based on Controllable Beam Squint

We further propose a high-accuracy localization scheme based on controllable beam squint. Specifically, the BS needs to implement P times TTDs-assisted beamforming to sense the positions of all users. In the p -th beam sweeping, by adjusting the values of PSs and TTDs, we can focus the beamforming of the 0-th and the M -th subcarriers on positions $(r_{mid1}, \theta_{max,p})$ and $(r_{mid2}, \theta_{min,p})$, respectively. In addition, we request that $\min\{\theta_{max,1}, \theta_{max,2}, \dots, \theta_{max,P}\} \geq \theta_{max}$, $\max\{\theta_{min,1}, \theta_{min,2}, \dots, \theta_{min,P}\} \leq \theta_{min}$, $\theta_{max,i} \neq \theta_{max,j}$ and $\theta_{min,i} \neq \theta_{min,j}$ for $i \neq j$, such that every time beam sweeping can cover the required angle sensing range, but the range of each beam sweeping is slightly different.

Based on formula (22), we can easily represent the received signal after being processed on the m -th subcarrier of the k -th user in the p -th beam sweeping as

$$\check{y}_{k,m,p}^{Pro2} = \frac{1}{r_k} e^{j\xi_{0,k,m,p}^{Pro2}} \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m,p}^{Pro2}} e^{jn^2\xi_{2,k,m,p}^{Pro2}}, \quad (25)$$

where $\xi_{0,k,m,p}^{Pro2} = \frac{2\pi}{Wc} [Wf_m r_k - (W - \tilde{f}_m) f_0 r_{mid1} - \tilde{f}_m f_M r_{mid2}]$, $\xi_{1,k,m,p}^{Pro2} = -\frac{2\pi d}{Wc} [Wf_m \sin \theta_k - (W - \tilde{f}_m) f_0 \sin \theta_{max,p} - \tilde{f}_m f_M \sin \theta_{min,p}]$ and $\xi_{2,k,m,p}^{Pro2} = \frac{2\pi d^2}{Wc} [Wf_m \frac{\cos^2 \theta_k}{2r_k} - (W - \tilde{f}_m) f_0 \frac{\cos^2 \theta_{max,p}}{2r_{mid1}} - \tilde{f}_m f_M \frac{\cos^2 \theta_{min,p}}{2r_{mid2}}]$. On the one hand, according to the analysis of the angle sensing stage in the proposed low complexity localization scheme, there must be one peak value in the received signal power spectrum of the k -th user in the p -th beam sweeping, whose corresponding maximum power subcarrier frequency can be recorded as $f_{k,d_\theta,p}^{Pro2}$, and the user's angle will be estimated as $\hat{\theta}_{k,p}^{Pro1}$. Besides, we mark the index of this subcarrier as $m_{k,p}$, and the theoretical phase of the k -th user's maximum power subcarrier in the p -th beam sweeping can be expressed as $\phi_{k,p}^{Pro2} = \arg\{\check{y}_{k,m_{k,p}}^{Pro2}\}$. On the other hand, users can detect the phase on each subcarrier of their received signal through the built-in channel detector. Record the phase measurement value of the k -th user's maximum power subcarrier in the p -th beam sweeping as $\varphi_{k,p}^{Pro2}$. Therefore, the k -th user can feedback $f_{k,d_\theta,p}^{Pro2}$ and $\varphi_{k,p}^{Pro2}$ to the BS.

After P times beam sweeping, the BS will receive the information about the k -th user as $\{f_{k,d_\theta,1}^{Pro2}, f_{k,d_\theta,2}^{Pro2}, \dots, f_{k,d_\theta,P}^{Pro2}\}$ and $\{\varphi_{k,1}^{Pro2}, \varphi_{k,2}^{Pro2}, \dots, \varphi_{k,P}^{Pro2}\}$, and BS can calculate the user's angle estimation result in each time beam sweeping as $\{\hat{\theta}_{k,1}^{Pro1}, \hat{\theta}_{k,2}^{Pro1}, \dots, \hat{\theta}_{k,P}^{Pro1}\}$. Then a higher accuracy angle

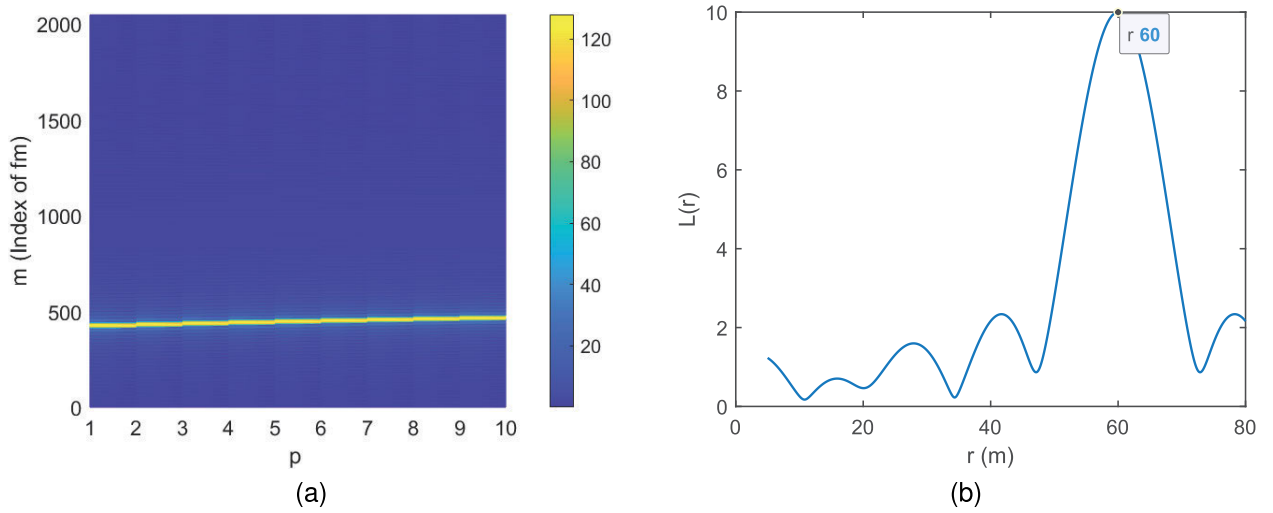


Fig. 9. An example of the proposed high-accuracy user localization scheme, in which the user is at $(60m, 30^\circ)$, $M = 2048$, $N = 128$ and $P = 10$. (a) The power spectrum matrix composed of the power spectrum received by the user during each beam sweeping. (b) Proposed one-dimensional distance search curve for user distance sensing.

estimation result for the k -th user can be obtained by taking the average value as

$$\hat{\theta}_k^{Pro2} = \frac{1}{P} \sum_{p=1}^P \hat{\theta}_{k,p}^{Pro1}. \quad (26)$$

In order to further improve the accuracy of distance estimation, we introduce the multiple carrier phase difference based ranging method. Due to the different phase changes caused by subcarriers of different frequencies transmitting the same distance, the transmission distance can be estimated based on the phase differences between multiple carriers [37], [38]. Due to the slightly different angle squint range set for each time beam sweeping, the frequency of the k -th user's maximum power subcarrier will vary during each beam sweeping. Hence the distance estimation result for the k -th user can be obtained through one-dimensional distance search as

$$\hat{r}_k^{Pro2} = \underset{r}{\operatorname{argmax}} L(r) = \underset{r}{\operatorname{argmax}} \left| \sum_{p=1}^P e^{j[\varphi_{k,p}^{Pro2} - \phi_{k,p,r}^{Pro2}]} \right|, \quad (27)$$

with

$$\begin{aligned} \phi_{k,p,r}^{Pro2} &= \arg\{\tilde{y}_{k,m'_{k,p},p,r}^{Pro2}\} \\ &= \arg\left\{ e^{j\xi_{0,k,m'_{k,p},p,r}^{Pro2}} \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{jn\xi_{1,k,m'_{k,p},p,r}^{Pro2}} e^{jn^2\xi_{2,k,m'_{k,p},p,r}^{Pro2}} \right\}, \end{aligned} \quad (28)$$

where $\xi_{1,k,m'_{k,p},p,r}^{Pro2} = -\frac{2\pi d}{Wc} [Wf_{k,d\theta,p}^{Pro2} \sin \hat{\theta}_{k,p}^{Pro1} - (W - \tilde{f}_{k,d\theta,p}^{Pro2})f_0 \sin \theta_{max,p} - \tilde{f}_{k,d\theta,p}^{Pro2}f_M \sin \theta_{min,p}] = 0$ (since $\hat{\theta}_{k,p}^{Pro1}$ is calculated by $f_{k,d\theta,p}^{Pro2}$ and (19)), $\xi_{0,k,m'_{k,p},p,r}^{Pro2} = \frac{2\pi}{Wc} [Wf_{k,d\theta,p}^{Pro2}r - (W - \tilde{f}_{k,d\theta,p}^{Pro2})f_0r_{mid1} - \tilde{f}_{k,d\theta,p}^{Pro2}f_Mr_{mid2}]$, $\xi_{2,k,m'_{k,p},p,r}^{Pro2} = \frac{2\pi d^2}{Wc} [Wf_{k,d\theta,p}^{Pro2} \frac{\cos^2 \hat{\theta}_{k,p}^{Pro1}}{2r} - (W - \tilde{f}_{k,d\theta,p}^{Pro2})f_0 \frac{\cos^2 \theta_{max,p}}{2r_{mid1}} - \tilde{f}_{k,d\theta,p}^{Pro2}f_M \frac{\cos^2 \theta_{min,p}}{2r_{mid2}}]$.

Equation (27) computes the difference between the measured phases and the theoretical phases over distance in peak power subcarriers. Essentially, the optimal distance estimation $\hat{r}_k^{Pro2}$ leads to the closest match between these two sets of phase values. Therefore, we can estimate the distance of the user by plotting r - $L(r)$ curve and searching for the peak.

An example of the proposed high-accuracy user localization scheme is shown in Fig. 9, where the user is located at $(60m, 30^\circ)$. Fig. 9 (a) shows the power spectrum matrix composed of the received signal power spectrum of the user in each time beam sweeping. It can be seen that the index of the maximum power subcarrier in the user's received signal is slightly different during each beam sweeping, and the maximum power subcarrier approximately forms a clear high power line in the entire spectrum matrix. Using the phase measurement values of the maximum power subcarriers with different frequencies, Fig. 9 (b) shows the one-dimensional distance search curve based on (27), in which an obvious peak corresponding to the search distance $60m$ can be found. Hence we can estimate the distance of the user as $60m$.

D. Proposed: Joint Localization Scheme Based on Double BSs and Controllable Beam Squint

The proposed high-accuracy localization scheme requires the one-dimensional distance search for each user during distance sensing, which may result in significant computational complexity. In order to reduce the computational complexity while also reducing the cost of beam sweeping times, we consider employing two BSs for user localization in ISAC system. The structure and principle of the double BSs localization system are shown in Fig. 10. It is assumed that both the two BSs are equipped with the ULA of N antennas, and the setting of the main BS is completely the same as the system model in Section II. The newly added auxiliary BS is parallel to the main BS, and its array center is located at $(L, 0)$. Other parameters of the auxiliary BS are identical to those of the main BS.

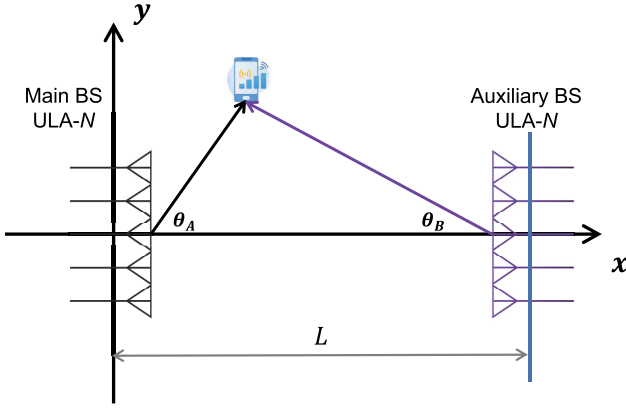


Fig. 10. Double BSs system.

 TABLE I
LIST OF ABBREVIATED NAMES FOR THE SCHEMES

Scheme	Abbreviation
Baseline: Traditional Beam Training Based Near-Field User Localization Scheme	TBT-Baseline
Proposed: A Low Complexity User Localization Scheme Based on Controllable Beam Squint	CBS-Low
Proposed: A High Accuracy User Localization Scheme Based on Controllable Beam Squint	CBS-High
Proposed: Joint Localization Scheme Based on Double BSs and Controllable Beam Squint	CBS-2BS

 TABLE II
OVERHEAD ANALYSIS

Scheme	Beam sweeping times for all users localization	Number of feedback values per user	Computing times per user	Number of BSs
TBT-Baseline	$I = I^a + K \times I^d$	2	2	1
CBS-Low	$K + 1$	2	2	1
CBS-High	P	$2P$	$P_R + P + 1$	1
CBS-2BS	2	2	3	2

For the k -th user, both the two BSs can use the angle sensing stage proposed in the low complexity localization scheme to obtain the user's angles with respect to each BS, which are denoted as $\theta_{A,k}$ and $\theta_{B,k}$ here. Then the angle and distance estimation results of the k -th user can be calculated according to the geometric relationship and trigonometric operation as

$$\hat{\theta}_k^{Pro3} = \theta_{A,k}, \quad (29)$$

$$\hat{r}_k^{Pro3} = \frac{L}{\cos \theta_{A,k} + \frac{\sin \theta_{A,k}}{\tan \theta_{B,k}}}. \quad (30)$$

E. Resource Overhead Analysis

We define the abbreviations of the four localization schemes involved in this paper as shown in Table I. In addition, the resource overhead of the four schemes are summarized in Table II, where P_R is the number of search grids for distance search in CBS-High Scheme.

In terms of beam sweeping cost, the localization schemes based on controllable beam squint can scan $M + 1$ near-field

positions in one time beam sweeping, while the TBT-Baseline Scheme can only scan one near-field position in one time beam sweeping. In order to ensure that the spatial scanning interval between the TBT-Baseline Scheme and the CBS Schemes is basically consistent, we set $I^a = M + 1$ and $I^d = M + 1$, respectively. Then the TBT-Baseline Scheme requires a total of $I = (K + 1)(M + 1)$ times beam sweeping for all users localization. In specific practice, there is $I \gg K + 1$, $I \gg P$ and $I \gg 2$. (More detailed numerical results can be found in the simulation section.) This means that the proposed beam squint based CBS-Low Scheme, CBS-High Scheme and CBS-2BS Scheme greatly reduce the overhead of beam sweeping times for sensing, which mainly due to the fact that the controllable beam squint can cover multiple near-field positions within only one OFDM symbol. Besides, the CBS-High Scheme improves the localization accuracy at the cost of computational complexity. The proposed CBS-2BS Scheme realize high-precision localization with low computational complexity and minimal beam sweeping times at the cost of adding an additional auxiliary BS.

In terms of user feedback cost, the localization schemes mentioned in this paper belong to cooperative user sensing schemes based on downlink detection and uplink feedback, both of which require user feedback. The TBT-Baseline Scheme requires each user to provide a total of 2 maximum power indices for feedback during the angle and distance estimation process. The CBS-Low Scheme requires each user to provide 2 frequency values of the maximum power subcarriers for feedback during the angle and distance sensing stages. The CBS-2BS Scheme requires each user to provide feedback with a total of 2 frequency values of the maximum power subcarriers for double BSs angles sensing. This means that the feedback costs of these three schemes are consistent. In addition, each user needs to provide feedback to the BS with P frequencies value and P phase values in CBS-High Scheme. However, we can see in the simulation that P is usually not set very large, and its typical value is below 10. Therefore, we believe that compared to the significantly reduced sensing beam sweeping times overhead and higher localization accuracy brought by CBS-High Scheme, the introduction of $2P - 2$ additional user feedback values can be accepted.

V. SIMULATION RESULTS

In simulations, root mean square error (RMSE) is employed to characterize the sensing performance. The RMSE of angle sensing, distance sensing and 2D localization can be defined as

$$\text{RMSE}_\theta = \sqrt{\frac{\sum_{t=1}^T (\hat{\theta}_{sp,t} - \theta_{sp})^2}{T}}, \quad \text{RMSE}_r = \sqrt{\frac{\sum_{t=1}^T (\hat{r}_{sp,t} - r_{sp})^2}{T}}$$

and $\text{RMSE}_{2D} = \sqrt{\frac{\sum_{t=1}^T [(\hat{x}_{sp,t} - x_{sp})^2 + (\hat{y}_{sp,t} - y_{sp})^2]}{T}}$, where T is the number of repeated experiments, (r_{sp}, θ_{sp}) and (x_{sp}, y_{sp}) are the user's real position, and $(\hat{r}_{sp,t}, \hat{\theta}_{sp,t})$ and $(\hat{x}_{sp,t}, \hat{y}_{sp,t})$ are the estimations of the user's position. Assume that the received signal on the m -th subcarrier is $y_m^{\text{noise}} = y_m^{\text{no}} + n_m^{\text{noise}}$, where y_m^{no} is a signal without noise, n_m^{noise} is the zero-mean additive complex Gaussian white noise, and y_m^{noise} is the signal after adding noise. Then the signal-to-noise ratio (SNR) can be defined as $\text{SNR} = \frac{\mathbb{E}\{|y_m^{\text{no}}|^2\}}{\mathbb{E}\{|n_m^{\text{noise}}|^2\}}$.

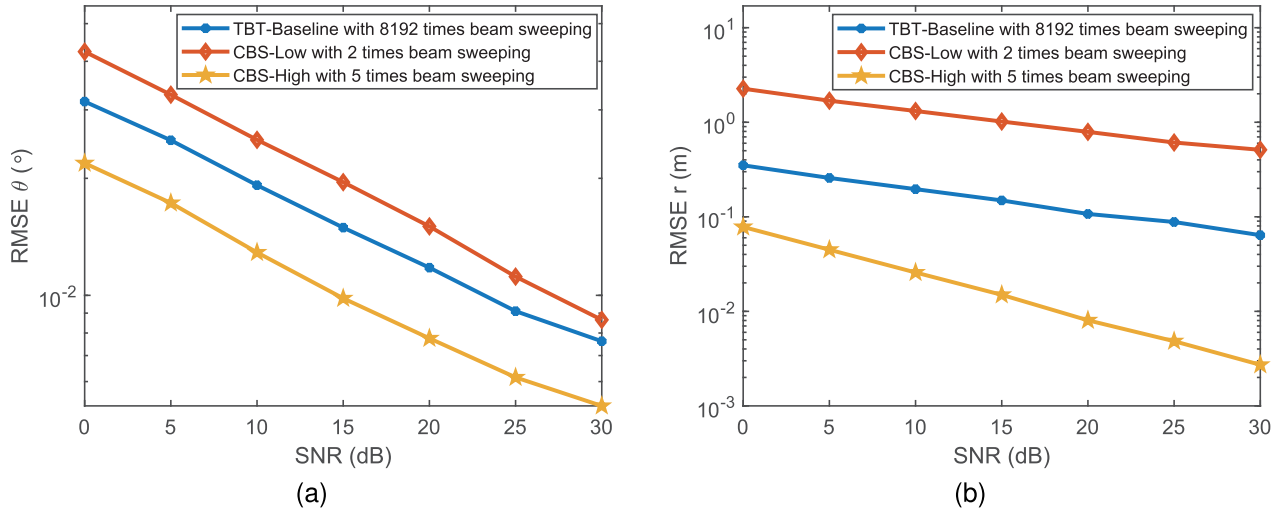


Fig. 11. (a) Angle sensing RMSE of three single BS localization schemes. (b) Distance sensing RMSE of three single BS localization schemes. Note that different schemes require different beam sweeping times.

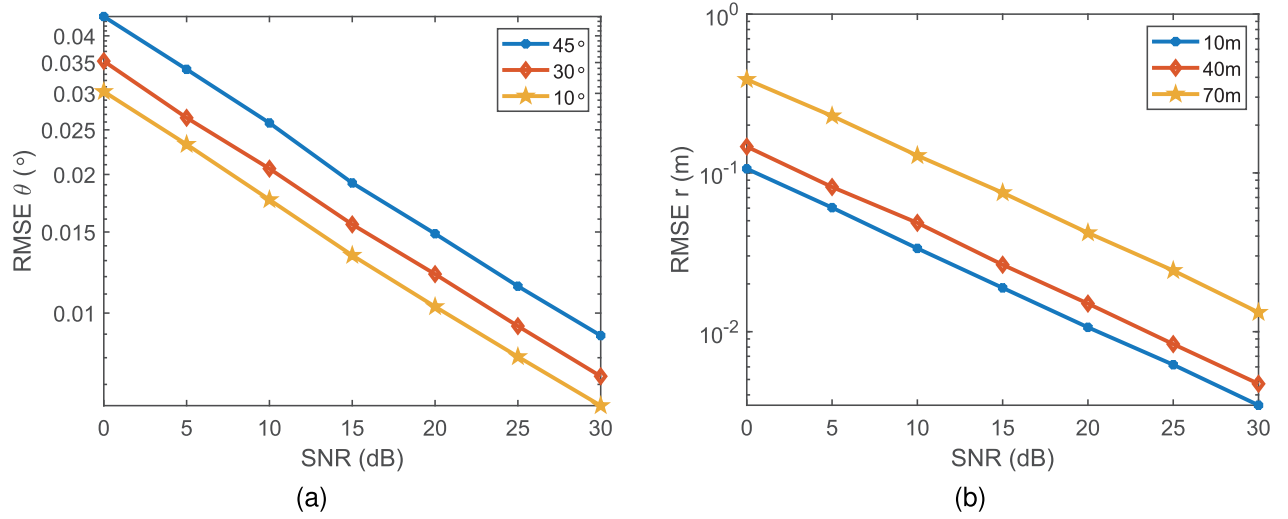


Fig. 12. (a) Angle sensing RMSE of the proposed CBS-High Scheme. (b) Distance sensing RMSE of the proposed CBS-High Scheme. Different curves represent users at different locations.

A. Localization Performance Based on Single BS

The TBT-Baseline Scheme, proposed CBS-Low Scheme, and proposed CBS-High Scheme are the localization strategies based on single BS. In the simulation of this subsection, the number of BS antennas is $N = 256$, the lowest carrier frequency is $f_0 = 60$ GHz, the OFDM bandwidth is $W = 4$ GHz, the number of subcarriers is $M + 1 = 4096$ and the antenna spacing is $d = \frac{1}{2}\lambda = \frac{c}{2f_0}$. Considering the Electromagnetic and Rayleigh distances, the sensing range of BS system is set to $\{(r, \theta) | 5m \leq r \leq 50m, -60^\circ \leq \theta \leq 60^\circ\}$. Then the average angle and distance intervals of controllable beam squint within the sensing range are $\frac{120^\circ}{4096-1} = 0.029^\circ$ and $\frac{45m}{4096-1} = 0.011m$, respectively. To compare these schemes more fairly, we set $I^a = M + 1 = 4096$ and $I^d = M + 1 = 4096$ in TBT-Baseline Scheme. This setting can ensure that the spatial scanning interval between TBT-Baseline Scheme and controllable beam squint is basically consistent.

Fig. 11 shows the curves of the localization RMSE with SNR for three Schemes. Under such parameter settings, the

TBT-Baseline Scheme requires $4096 + 4096 = 8192$ times beam sweeping to complete the localization for the first user, and additional 4096 times beam sweeping are required for each additional user. The proposed CBS-Low Scheme only needs 2 times beam sweeping to obtain the position of the first user, additional one time beam sweeping is required for each additional user, and its localization performance is only slightly lower than the TBT-Baseline Scheme. This means that the CBS-Low Scheme can realize acceptable user localization accuracy while saves 99.9755% of the beam sweeping cost compared to the TBT-Baseline Scheme, thanks to the wide coverage characteristics of beam squint. In addition, it can be seen that the RMSE of the CBS-High Scheme with only $P = 5$ times beam sweeping is much lower than that of TBT-Baseline Scheme with 8192 times beam sweeping. And the CBS-High Scheme does not require additional beam sweeping for additional user localization. This means that the beam sweeping speed of the CBS-High Scheme is at least 1638 times faster than that of TBT-Baseline Scheme. The simulation

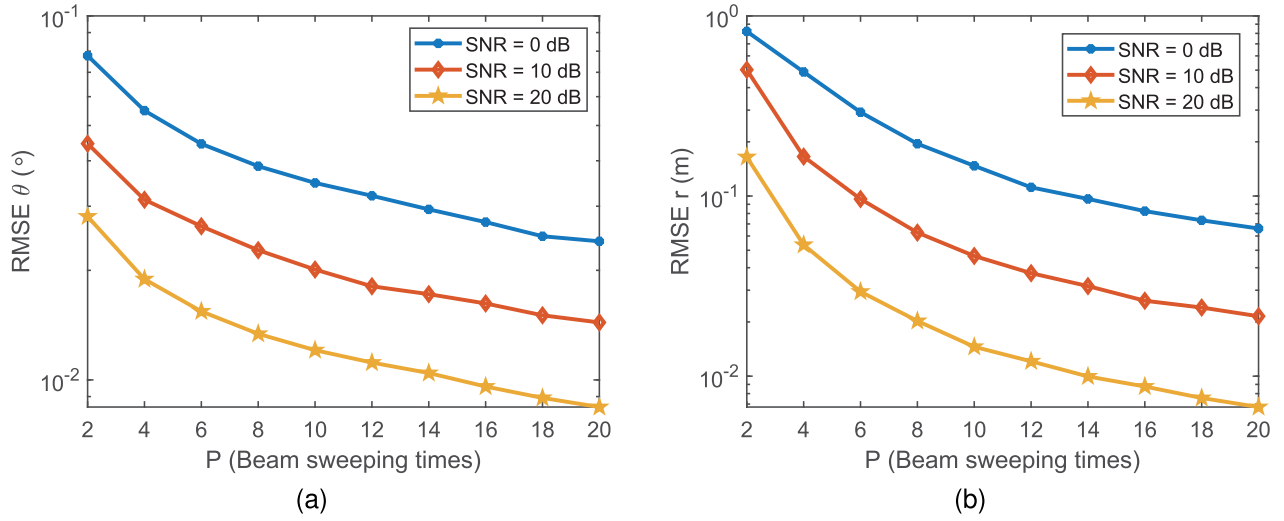


Fig. 13. (a) The angle sensing RMSE of the proposed CBS-High Scheme with different P . (b) The distance sensing RMSE of the proposed CBS-High Scheme with different P .

results indicate that the frequency-domain beam sweeping based on controllable beam squint can replace traditional time-domain beam sweeping, which can greatly reduce the time cost of beam sweeping for sensing.

B. Localization Performance of Users at Different Positions in CBS-High Scheme

Here, we will demonstrate the localization performance of the proposed CBS-High Scheme for users at different locations. Assume the number of BS antennas is $N = 128$, the carrier frequency is $f_0 = 60$ GHz, the transmission bandwidth is $W = 1$ GHz, and the number of subcarriers is $M + 1 = 2048$. The sensing range of BS system is set to $\{(r, \theta) | 5m \leq r \leq 80m, -60^\circ \leq \theta \leq 60^\circ\}$. Fig. 12 shows the RMSE for locating users at different positions, in which both $RMSE_\theta$ and $RMSE_r$ decrease with the increase of SNR.

The three users in Fig. 12 (a) are at the same distance but different angles. It can be found that the closer the user's angle is to 0° , the lower the $RMSE_\theta$ will be. This phenomenon is mainly due to the fact that controllable beam squint gathers more subcarriers near 0° in angle sensing stage, which can be inferred from (19), resulting in better angle sensing performance near 0° . Besides, the three users in Fig. 12 (b) are at the same angle but different distances. It shows that the users who are closer to the BS have higher distance sensing accuracy, which is due to the fact that near-field beamforming has better focusing performance at closer distances.

C. The Impact of Beam Sweeping Times on Localization Performance in CBS-High Scheme

The CBS-High Scheme requires the implementation of P times TTDs-assisted beamforming with different angle squint ranges to realize user localization. Assuming the number of BS antennas is $N = 128$, the number of subcarriers is $M + 1 = 2048$ and the user is at $(60m, 20^\circ)$. Fig. 13 shows the variation curve of the localization RMSE with the number of beam sweeping times P under a fixed SNR. It can be

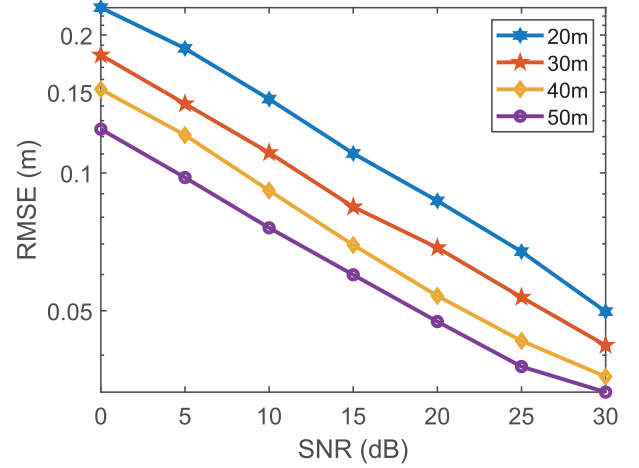


Fig. 14. The 2D localization RMSE of the proposed double BSs and beam squint based user localization scheme. The users' angles are 30° and users' distances are different. $L = 100m$, $N = 256$ and $M + 1 = 2048$.

seen that both the $RMSE_\theta$ and $RMSE_r$ will decrease as the increase of P . When $P = 4$ and SNR = 10 dB, the $RMSE_\theta$ is 0.031° and the $RMSE_r$ is $0.165m$. When P increases to 12, the $RMSE_\theta$ decreases to 0.018° and the $RMSE_r$ decreases to $0.037m$. Besides, increasing P also requires more time to complete sensing. Therefore, we can balance between the sensing accuracy and the acceptable beam sweeping time overhead.

D. Simulation of Double BSs and Beam Squint Based Localization Scheme

In the simulation of double BSs and beam squint based localization Scheme, the number of BS antennas is set as $N = 256$, the number of subcarriers is $M + 1 = 2048$, and the distance between the two BSs is $L = 100m$. The 2D localization RMSE of the proposed CBS-2BS Scheme is shown in Fig. 14. It can be seen that the average 2D RMSE of all users is approximately $0.17m$ when SNR is 0 dB, and gradually decrease to $0.10m$ when SNR increases to 15 dB.

This indicates that the proposed CBS-2BS Scheme can obtain the high-precision position estimations for all users with the cost of only two times beam sweeping. In addition, the 2D RMSE of the users whose abscissa (x_{sp}) is close to $\frac{L}{2}$ will be smaller, which is caused by geometric symmetry.

VI. CONCLUSION

This paper proposes the low-overhead user localization schemes based on beam squint in near-field ISAC system. With the aid of the TTDs, the range and trajectory of the beam squint can be freely controlled, and hence it is possible to reversely utilize the beam squint for user localization. Then we derive the trajectory equation for near-field beam squint points and design a way to control the trajectory of these beam squint points. With the proposed design, beamforming from different subcarriers would purposely point to different angles and different distances such that users from different positions would receive the maximum power at different subcarriers. Hence, one can simply obtain the different users' position from the beam squint effect. Furthermore, we utilize the phase difference of the maximum power subcarriers received by the user at different frequencies in several times beam sweeping to obtain a more accurate distance estimation result, ultimately realizing high accuracy and low beam sweeping overhead user localization. Simulation results demonstrate the effectiveness of the proposed schemes.

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